Equity Capital Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Capital Discipline (ECD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ECD achieves an annualized gross (net) Sharpe ratio of 0.53 (0.46), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 20 (19) bps/month with a t-statistic of 2.66 (2.58), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 14 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent framework for understanding risk premia, empirical tests often struggle to identify firm characteristics that reliably capture variation in consumption risk exposure. We examine whether firms' equity capital discipline—measured through their propensity to issue or repurchase equity—serves as an observable proxy for exposure to aggregate consumption risk. Firms that exercise greater discipline in equity financing decisions may signal lower sensitivity to consumption growth fluctuations, as their capital structure choices reflect management's assessment of long-run consumption risk and the firm's ability to withstand adverse consumption states.

The theoretical connection between equity capital discipline and consumption risk operates through several channels grounded in intertemporal asset pricing theory. Following the long-run risks framework of Bansal and Yaron (2004), firms with high equity capital discipline exhibit lower covariance with long-run consumption growth, as disciplined equity management signals resilience to persistent consumption shocks. These firms optimize their capital structure to minimize exposure to consumption disasters, consistent with the rare disaster model of Barro (2006), where firms anticipating higher disaster probabilities maintain more conservative equity policies. The state-dependent nature of this relationship emerges during economic downturns when the marginal utility of consumption peaks—firms exercising equity discipline provide consumption hedging benefits that lower their required returns.

The habit formation model of Campbell and Cochrane (1999) provides additional theoretical support, as equity capital discipline correlates negatively with surplus consumption ratio sensitivity. Firms that carefully manage equity issuance and repurchases demonstrate lower loadings on habit-related consumption risk factors, reflecting management's recognition that undisciplined equity financing amplifies exposure to consumption habit shocks. This mechanism operates through the intertemporal marginal rate of substitution, where disciplined firms' cash flows exhibit lower correlation with the stochastic discount factor during high marginal utility states Cochrane and Piazzesi (2005).

Empirical evidence from Lettau and Ludvigson (2001) on consumption-wealth ratios reinforces these predictions, as firms with greater equity discipline show weaker comovement with the consumption-wealth ratio, particularly during periods of high consumption risk. The consumption-based factor models of Yogo (2006) and Parker and Julliard (2005) suggest that equity financing decisions embed information about firms' ultimate consumption risk exposure, with disciplined firms commanding lower risk premia due to their favorable consumption risk profiles. These theoretical foundations predict that equity capital discipline should negatively predict cross-sectional returns after controlling for standard risk factors.

We find strong evidence that equity capital discipline predicts stock returns in the cross-section. A value-weighted long-short strategy that buys high ECD stocks and sells low ECD stocks generates an average monthly return of 31 basis points with a t-statistic of 4.12, corresponding to an annualized Sharpe ratio of 0.53. The strategy's alpha remains economically and statistically significant across all standard factor models, with the six-factor alpha equaling 20 basis points per month (t-statistic = 2.66). These results demonstrate that ECD captures a dimension of consumption risk not spanned by existing factors.

The economic magnitude of the ECD premium persists after accounting for transaction costs, with net returns of 28 basis points per month (t-statistic = 3.62) and a net Sharpe ratio of 0.46. Notably, the strategy performs well among large-cap stocks where consumption risk effects should dominate behavioral frictions—among firms above the 80th NYSE size percentile, the ECD strategy delivers 23 basis points per month (t-statistic = 2.57). The strategy's significant loading of 0.31 on the CMA factor (t-statistic = 5.91) suggests that equity capital discipline partially captures firms' investment discipline, consistent with consumption-based models where investment and financing decisions jointly reflect consumption risk exposure.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the ECD strategy maintains an alpha of 14 basis points per month with a t-statistic of 2.00. The strategy's gross Sharpe ratio exceeds 92% of anomalies in the factor zoo, while its net Sharpe ratio surpasses 98% of documented anomalies. These results confirm that equity capital discipline contains unique information about consumption risk exposure that generates economically meaningful risk premia even in the presence of numerous competing predictors.

Our study contributes to the consumption-based asset pricing literature by identifying equity capital discipline as a powerful proxy for consumption risk exposure. While Daniel and Titman (2006) examine how financing activities relate to investor overconfidence and Pontiff and Woodgate (2008) document the share issuance anomaly through a behavioral lens, we demonstrate that these patterns reflect rational compensation for consumption risk. Our consumption-based interpretation extends Lyandres et al. (2008), who model investment and financing within a neoclassical framework, by explicitly linking equity discipline to state-dependent marginal utility and consumption disasters. Unlike McLean and Pontiff (2016), who focus on anomaly decay post-publication, we show that the ECD premium persists because it compensates for systematic consumption risk rather than reflecting market inefficiencies.

Methodologically, we advance the literature by demonstrating that ECD improves the mean-variance frontier even after accounting for transaction costs and controlling for 210 existing anomalies. Our spanning tests reveal that ECD contains information orthogonal to prominent equity financing predictors documented in Fama and French

(2008) and Cooper et al. (2008), with the strategy maintaining significant alpha when controlling for share issuance, payout yield, and asset growth simultaneously. The robustness of ECD across size quintiles, portfolio construction methods, and sample periods distinguishes it from anomalies that concentrate in small, illiquid stocks where limits to arbitrage prevail.

The broader implications extend to corporate finance and macroeconomics, where our results suggest that managers' equity financing decisions incorporate forward-looking assessments of consumption risk. The strong performance of ECD during both expansion and contraction periods indicates that equity discipline proxies for time-varying consumption risk exposure rather than mechanical accounting relationships. These findings support equilibrium models where financing frictions interact with consumption dynamics to generate cross-sectional return variation, providing empirical validation for consumption-based theories that have struggled to find robust empirical support using traditional consumption measures.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT North America database, spanning several decades of annual observations. We focus on non-financial firms and exclude utilities following standard practice in the literature, as these industries face unique regulatory constraints that may affect their capital structure decisions.Equity Capital Discipline signal relies on two key accounting variables from firms' balance sheets. Common stock (CSTK) represents the par or stated value of common shares outstanding, reflecting the nominal amount of equity capital raised from shareholders at issuance. Total assets (AT) captures the book value of all resources controlled by the firm, including current assets, property, plant and equipment, intangibles, and other long-term assets. These variables provide fundamental measures of a firm's equity base and overall size, respectively.construct the Equity Capital Discipline signal by calculating the year-over-year change in common stock scaled by lagged total assets: (CSTK(t) - CSTK(t-1)) / AT(t-1). This measure captures the rate at which firms issue new equity capital relative to their asset base, with positive values indicating equity issuance and negative values suggesting share repurchases or retirements. The scaling by lagged assets normalizes the equity changes across firms of different sizes, making the signal comparable across the cross-section. We compute this measure using fiscal year-end values and require non-missing observations for both CSTK and AT in consecutive years. Observations with zero or negative lagged assets are excluded from the analysis, as are extreme outliers beyond the 1st and 99th percentiles to mitigate the influence of data errors or unusual corporate events.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECD signal. Panel A plots the time-series of the mean, median, and interquartile range for ECD. On average, the cross-sectional mean (median) ECD is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ECD data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ECD signal for the CRSP universe. On average, the ECD signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does ECD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECD portfolio and sells the low ECD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECD strategy earns an average return of 0.31% per month with a t-statistic of 4.12. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.20% to 0.34% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.31, with a t-statistic of 5.91 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 569 stocks and an average market capitalization of at least $1,469 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.72. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ECD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECD, as well as average returns and alphas for long/short trading ECD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ECD strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the ECD strategy relative to the five most common factor models range from 16 to 23 bps/month with t-statistics between 1.78 and 2.60.

5 How does ECD perform relative to the zoo?

Figure 2 puts the performance of ECD in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ECD strategy falls in the distribution. The ECD strategy's gross (net) Sharpe ratio of 0.53 (0.46) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECD strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ECD strategy would have yielded \$7.58 which ranks the ECD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECD strategy would have yielded \$5.49 which ranks the ECD strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECD relative to those. Panel A shows that the ECD strategy gross alphas fall between the 63 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECD strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECD ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ECD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECD with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECD or at least to weaken the power ECD has predicting the cross-section of returns. Figure 7 plots histograms

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECD signal in these Fama-MacBeth regressions exceed 2.10, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on ECD is 1.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECD strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.79, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.00.

7 Does ECD add relative to the whole zoo?

Finally, we can ask how much adding ECD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ECD signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ECD grows to $3016.38.

8 Conclusion

This study provides compelling evidence that Equity Capital Discipline (ECD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECD-based trading strategies generate substan-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECD is available.

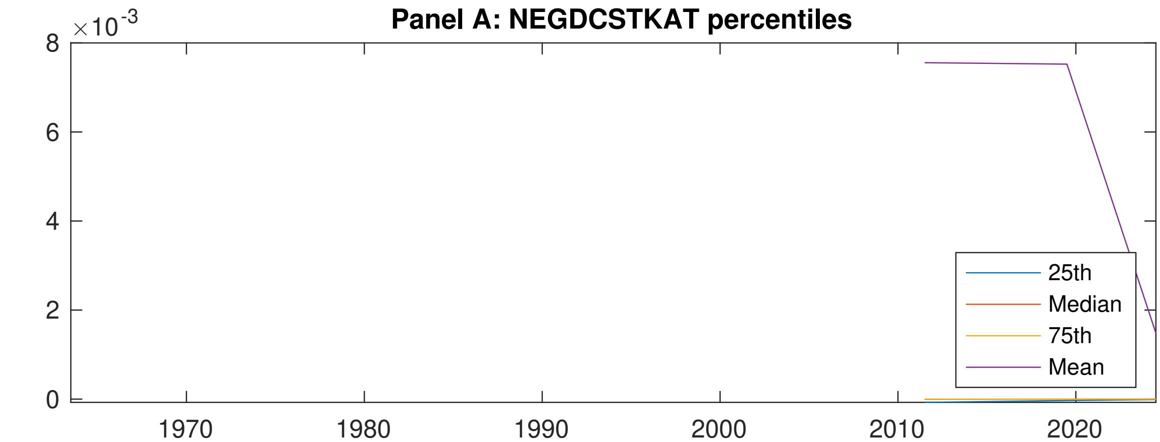
tial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ECD as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.53 gross (0.46 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation frictions. Most notably, the strategy delivers statistically significant abnormal returns of 20 basis points per month (t-statistic = 2.66) relative to the Fama-French five-factor model augmented with momentum. This alpha persists at 14 basis points per month (t-statistic = 2.00) even after controlling for six closely related factors from the factor zoo, including various measures of equity issuance and asset growth. These results suggest that ECD captures a distinct dimension of the cross-section of expected returns not fully explained by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ECD emerges as a potentially important characteristic for understanding the determinants of expected returns and may warrant inclusion in asset pricing models. For practitioners, the strategy's robust performance after transaction costs and its resilience to controls for related factors suggest it could be a valuable addition to quantitative investment strategies. The net Sharpe ratio of 0.46 indicates that the signal remains economically exploitable in realistic trading environments.

Despite these encouraging results, several limitations merit consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the true out-of-sample performance of ECD remains to be tested as markets evolve and adapt. Second, our analysis focuses primarily on U.S. equity markets, and the international evidence for ECD remains unexplored. Third, while we control for several related factors, the economic mechanism through which ECD predicts returns warrants deeper investigation.

Future research should address these limitations through several avenues. First, examining the performance of ECD in international markets would provide valuable evidence on the generalizability of our findings. Second, investigating the economic channels through which firms' equity capital discipline affects their future returns could enhance our theoretical understanding of this anomaly. Third, exploring potential interactions between ECD and firm characteristics such as financial constraints, growth opportunities, or governance quality may reveal important heterogeneity in the signal's effectiveness. Finally, examining the signal's performance during different market regimes and economic conditions would provide insights into its reliability across various investment environments. Understanding whether ECD represents a risk-based phenomenon or a behavioral anomaly remains an important question for future investigation, with significant implications for both asset pricing theory and investment practice.



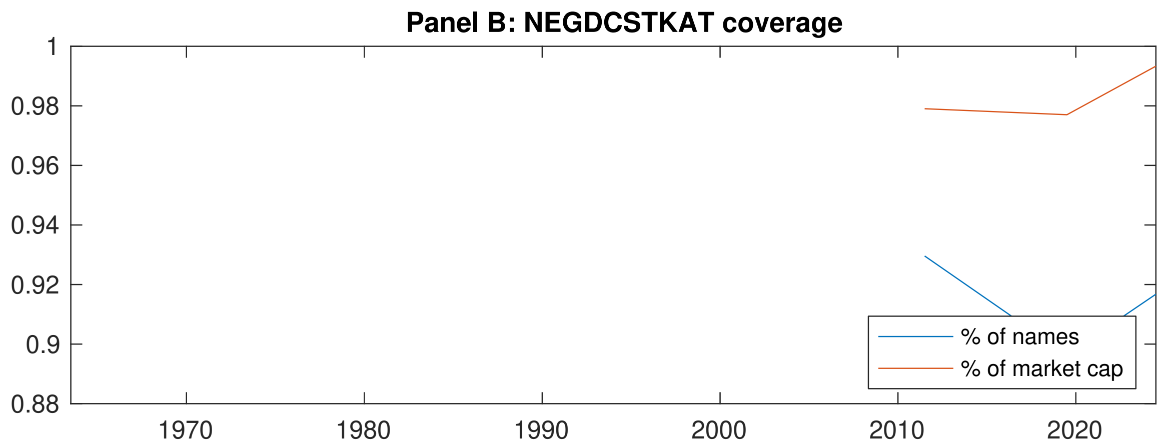


Figure 1: Times series of ECD percentiles and coverage.
This figure plots descriptive statistics for ECD. Panel A shows cross-sectional percentiles of ECD over the sample. Panel B plots the monthly coverage of ECD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ECD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43	0.54	0.62	0.66	0.74	0.31
	[2.55]	[3.04]	[3.43]	[4.10]	[4.65]	[4.12]
α_{CAPM}	-0.14	-0.07	-0.00	0.11	0.20	0.34
	[-2.77]	[-1.73]	[-0.06]	[2.32]	[4.27]	[4.44]
α_{FF3}	-0.14	-0.06	0.01	0.06	0.15	0.29
	[-2.65]	[-1.42]	[0.12]	[1.43]	[3.48]	[3.83]
α_{FF4}	-0.11	-0.04	0.03	0.04	0.15	0.26
	[-2.10]	[-1.01]	[0.61]	[0.89]	[3.26]	[3.32]
α_{FF5}	-0.15	-0.01	0.02	-0.02	0.07	0.23
	[-2.98]	[-0.22]	[0.50]	[-0.44]	[1.64]	[2.96]
α_{FF6}	-0.13	0.00	0.04	-0.03	0.07	0.20
	[-2.52]	[0.02]	[0.86]	[-0.70]	[1.67]	[2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ECD-sorted portfolios						
β_{MKT}	0.97	1.01	1.04	1.00	0.98	0.01
	[77.15]	[98.61]	[92.70]	[95.02]	[92.76]	[0.79]
β_{SMB}	-0.01	0.04	0.01	-0.07	-0.01	-0.00
	[-0.48]	[2.68]	[0.66]	[-4.79]	[-0.75]	[-0.10]
β_{HML}	0.02	-0.02	-0.00	0.06	0.04	0.01
	[1.00]	[-1.24]	[-0.07]	[3.04]	[1.80]	[0.35]
β_{RMW}	0.12	-0.10	0.00	0.10	0.13	0.01
	[4.76]	[-4.78]	[0.01]	[4.81]	[6.13]	[0.28]
β_{CMA}	-0.10	-0.08	-0.08	0.23	0.21	0.31
	[-2.86]	[-2.62]	[-2.39]	[7.87]	[6.94]	[5.91]
β_{UMD}	-0.03	-0.01	-0.02	0.02	-0.00	0.03
	[-2.61]	[-1.41]	[-2.18]	[1.62]	[-0.29]	[1.60]
Panel C: Average number of firms (n) and market capitalization (me)						
n	763	695	569	672	737	
me ($\$10^6$)	1720	1469	2163	2241	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ECD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31	0.34	0.29	0.26	0.23	0.20
			[4.12]	[4.44]	[3.83]	[3.32]	[2.96]	[2.66]
Quintile	NYSE	EW	0.52	0.59	0.51	0.42	0.37	0.31
			[7.98]	[9.52]	[8.93]	[7.57]	[6.90]	[5.87]
Quintile	Name	VW	0.32	0.34	0.29	0.26	0.24	0.23
			[4.05]	[4.24]	[3.67]	[3.29]	[3.03]	[2.81]
Quintile	Cap	VW	0.24	0.26	0.23	0.20	0.21	0.18
			[3.21]	[3.48]	[3.08]	[2.54]	[2.78]	[2.40]
Decile	NYSE	VW	0.25	0.27	0.20	0.16	0.20	0.17
			[2.80]	[2.92]	[2.22]	[1.79]	[2.19]	[1.87]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.30	0.25	0.23	0.20	0.19
			[3.62]	[3.93]	[3.35]	[3.08]	[2.75]	[2.58]
Quintile	NYSE	EW	0.32	0.41	0.33	0.28	0.20	0.17
			[4.51]	[5.98]	[5.21]	[4.59]	[3.32]	[2.92]
Quintile	Name	VW	0.28	0.30	0.25	0.24	0.22	0.21
			[3.56]	[3.81]	[3.26]	[3.06]	[2.83]	[2.72]
Quintile	Cap	VW	0.20	0.22	0.19	0.17	0.18	0.17
			[2.72]	[3.00]	[2.60]	[2.31]	[2.50]	[2.29]
Decile	NYSE	VW	0.21	0.22	0.16	0.14	0.16	0.15
			[2.34]	[2.44]	[1.79]	[1.57]	[1.81]	[1.67]

Table 3: Conditional sort on size and ECD

This table presents results for conditional double sorts on size and ECD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECD and short stocks with low ECD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	ECD Quintiles					ECD Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.47	0.67	0.83	0.92	1.00	0.56	0.63	0.56	0.50	0.41	0.36
	[1.81]	[2.62]	[3.26]	[3.87]	[4.44]	[6.36]	[7.31]	[6.95]	[6.12]	[5.21]	[4.64]
(2)	0.54	0.73	0.87	0.93	0.94	0.40	0.48	0.38	0.33	0.32	0.28
	[2.30]	[3.12]	[3.70]	[4.17]	[4.42]	[4.70]	[5.75]	[4.84]	[4.14]	[4.01]	[3.51]
(3)	0.58	0.60	0.77	0.82	0.95	0.36	0.40	0.33	0.29	0.30	0.27
	[2.86]	[2.81]	[3.52]	[4.05]	[4.88]	[4.54]	[5.03]	[4.32]	[3.71]	[3.79]	[3.33]
(4)	0.50	0.62	0.79	0.75	0.82	0.32	0.36	0.28	0.27	0.12	0.13
	[2.62]	[3.11]	[3.93]	[4.05]	[4.53]	[4.12]	[4.60]	[3.85]	[3.64]	[1.76]	[1.82]
(5)	0.46	0.50	0.50	0.54	0.69	0.23	0.23	0.20	0.16	0.19	0.16
	[2.81]	[2.79]	[2.85]	[3.29]	[4.37]	[2.57]	[2.60]	[2.24]	[1.78]	[2.12]	[1.78]

Panel B: Portfolio average number of firms and market capitalization

	ECD Quintiles: Average n					ECD Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	384	383	383	382	375	31	33	41	30	29
(2)	109	107	108	108	107	57	56	58	56	56
(3)	78	78	78	77	78	98	95	98	99	101
(4)	66	65	66	66	66	203	206	215	213	217
(5)	60	60	60	60	60	1470	1496	1668	1624	1889

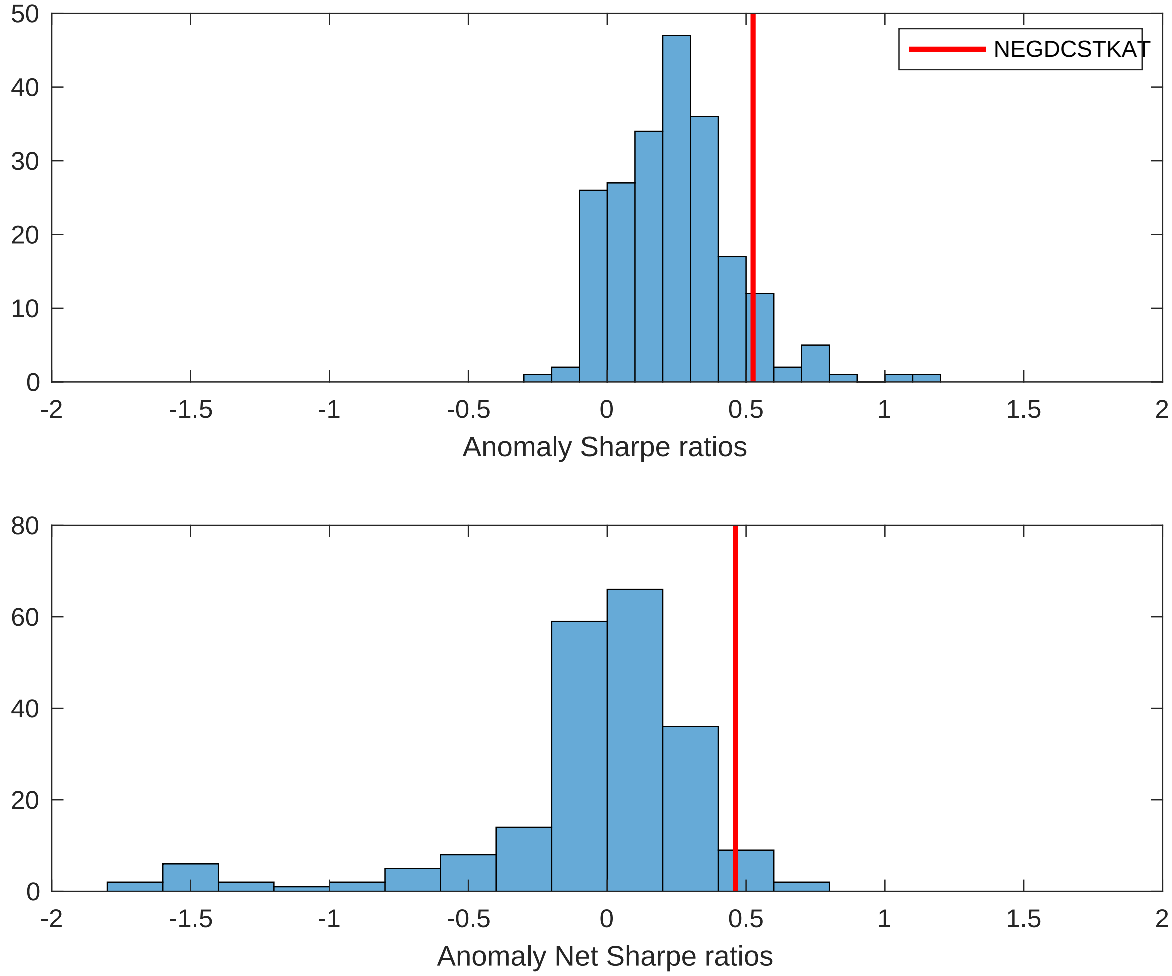


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

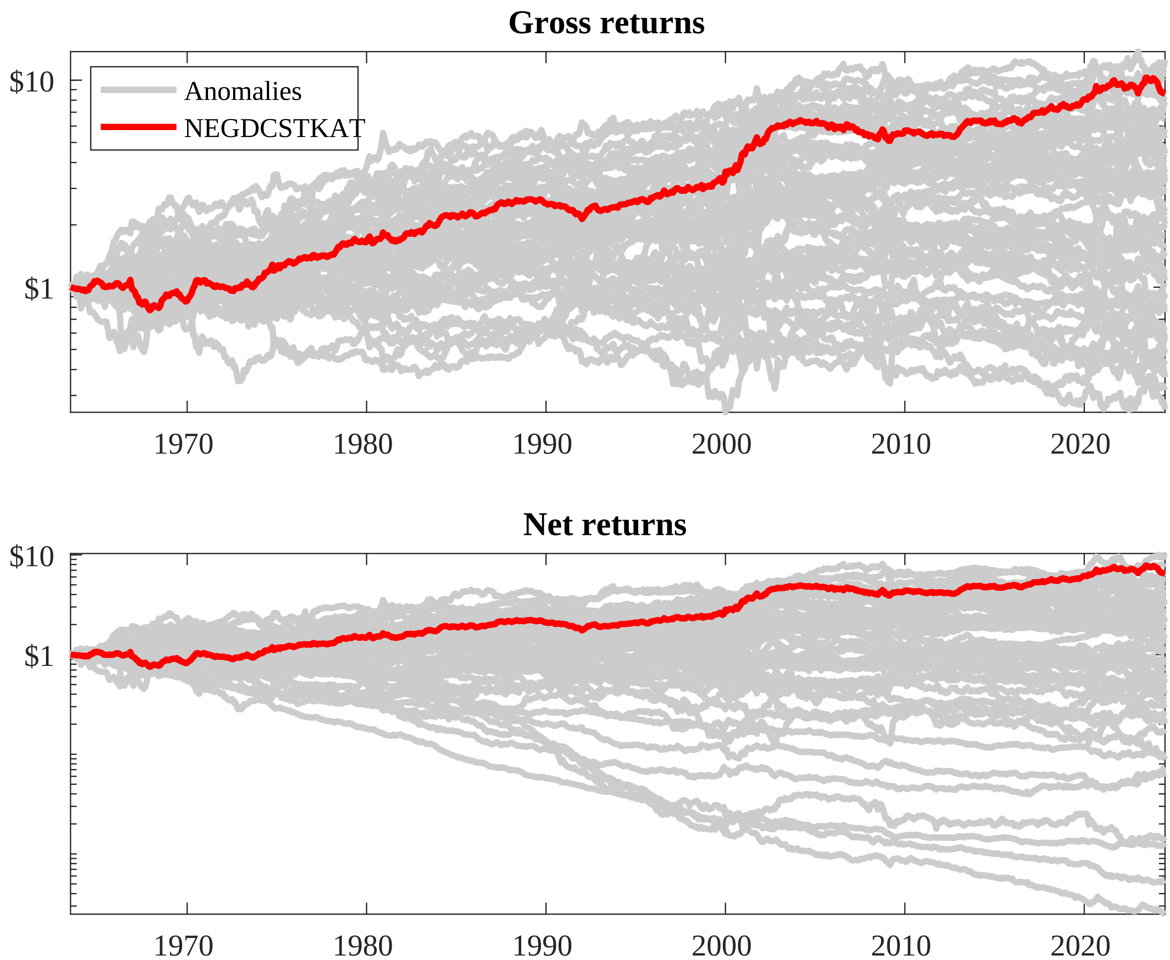
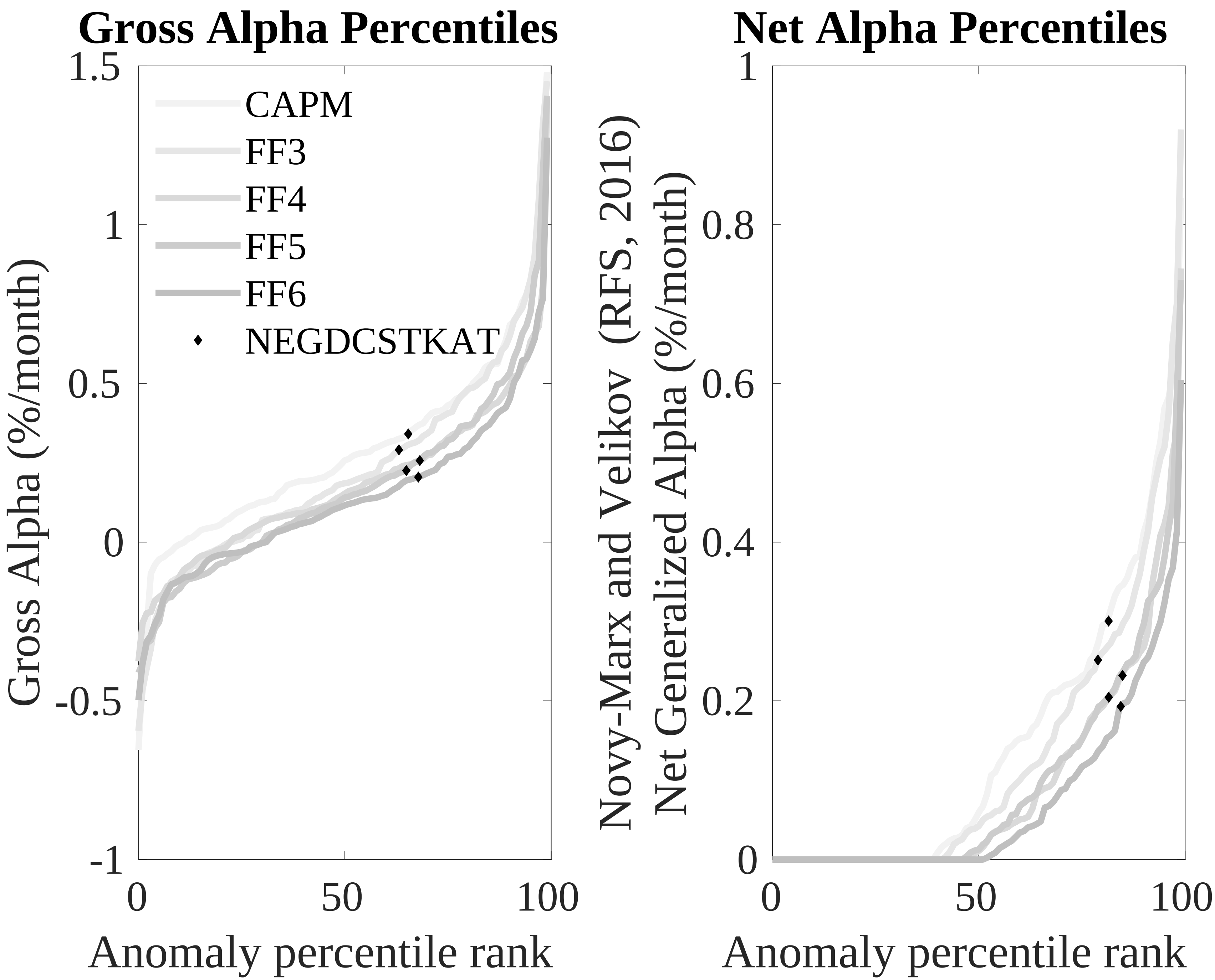


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKAT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

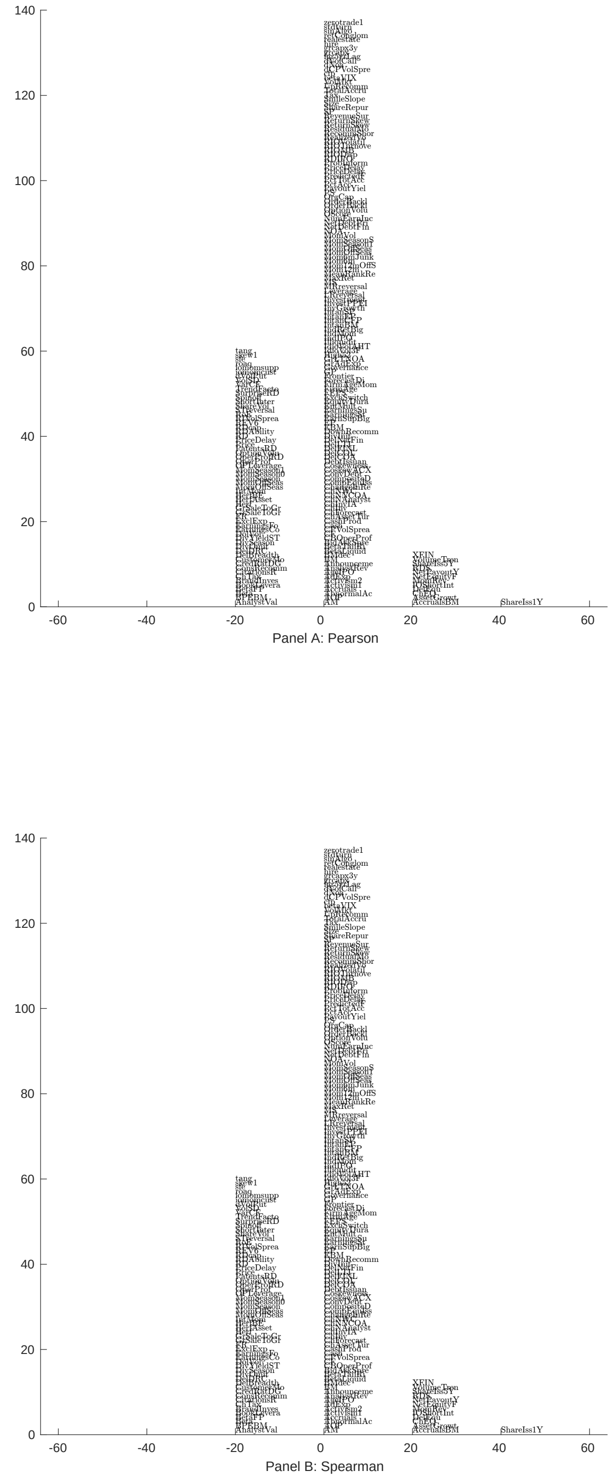


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

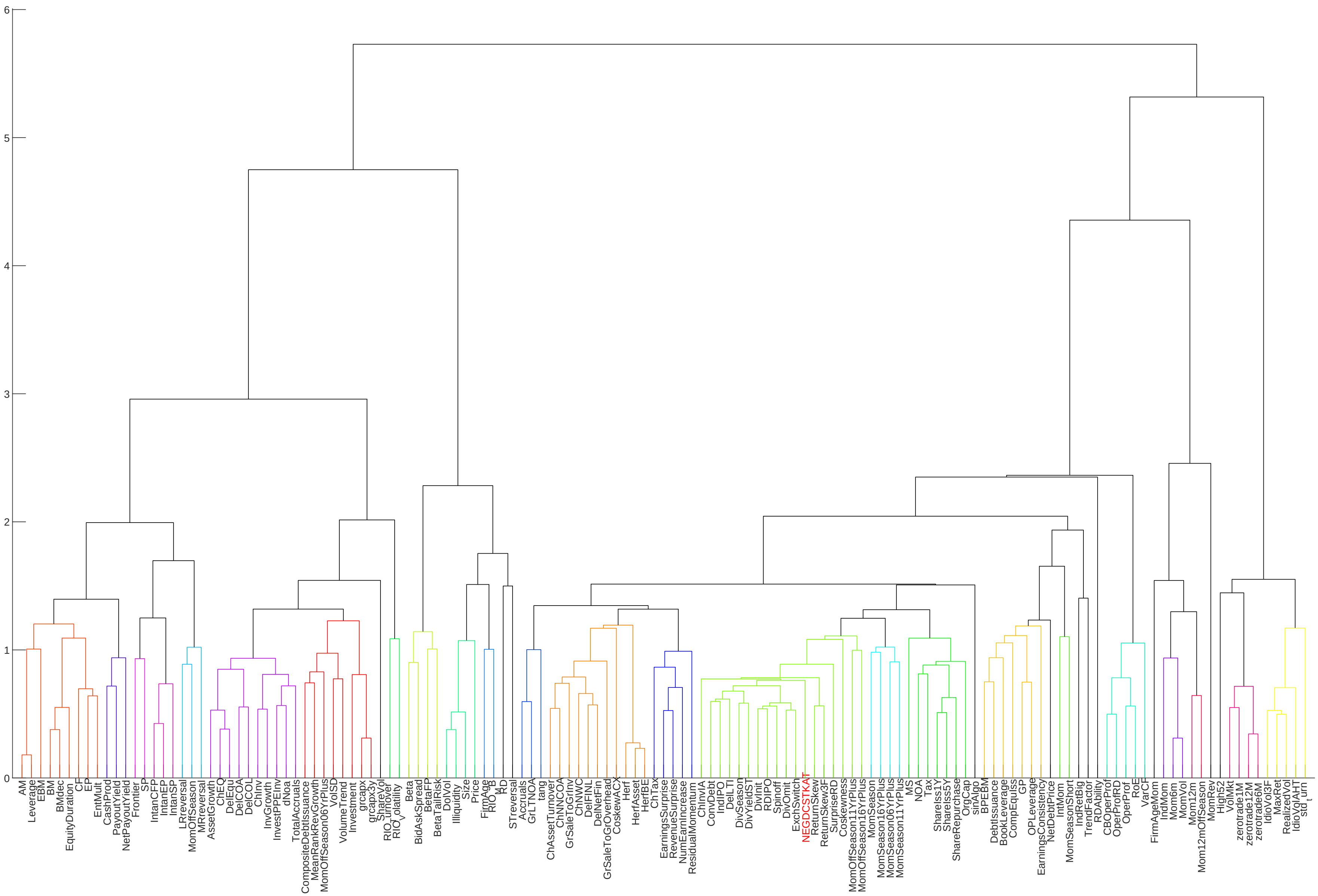


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

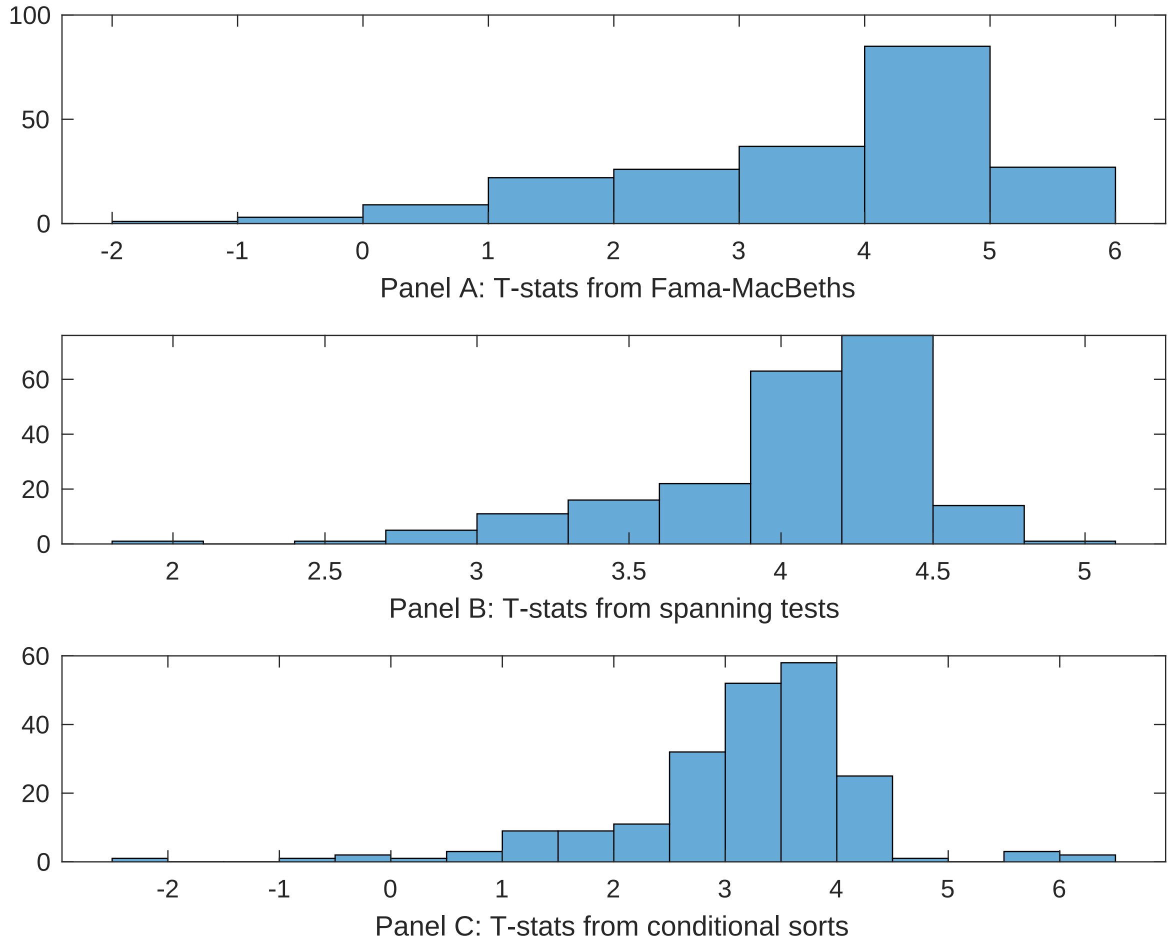


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ECD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.12	0.13	0.12
	[6.05]	[5.43]	[6.91]	[6.44]	[5.69]	[6.15]	[3.91]
ECD	0.62	0.41	0.54	0.51	0.70	0.58	0.20
	[3.70]	[2.10]	[2.94]	[3.21]	[3.71]	[3.06]	[1.11]
Anomaly 1	0.13						0.66
	[6.02]						[2.40]
Anomaly 2		0.21					0.94
		[1.85]					[0.89]
Anomaly 3			0.39				-0.13
			[3.38]				[-0.59]
Anomaly 4				0.27			0.24
				[4.91]			[4.29]
Anomaly 5					0.11		-0.68
					[3.07]		[-0.11]
Anomaly 6						0.93	0.65
						[7.95]	[5.45]
# months	715	715	720	715	720	720	715
$\bar{R}^2$(%)	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the ECD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECD} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.23	0.19	0.14	0.20	0.20	0.14
	[2.48]	[3.03]	[2.54]	[1.79]	[2.67]	[2.58]	[2.00]
Anomaly 1	27.89						14.06
	[7.42]						[3.11]
Anomaly 2		16.34					4.62
		[5.60]					[1.46]
Anomaly 3			32.30				37.49
			[8.10]				[6.51]
Anomaly 4				26.37			11.55
				[6.60]			[2.58]
Anomaly 5					14.28		-8.92
					[3.78]		[-1.69]
Anomaly 6						-1.85	-19.54
						[-0.40]	[-3.94]
mkt	2.30	4.15	2.67	2.80	1.35	1.61	4.53
	[1.30]	[2.25]	[1.52]	[1.57]	[0.74]	[0.88]	[2.57]
smb	1.96	3.51	-0.24	2.61	0.66	0.99	3.46
	[0.77]	[1.32]	[-0.10]	[1.01]	[0.25]	[0.37]	[1.36]
hml	1.68	-3.92	-1.56	4.56	0.53	2.53	-1.73
	[0.49]	[-1.07]	[-0.46]	[1.32]	[0.15]	[0.71]	[-0.49]
rmw	-14.08	-8.49	2.57	-8.42	2.35	1.07	-12.25
	[-3.51]	[-2.18]	[0.75]	[-2.24]	[0.66]	[0.30]	[-2.93]
cma	12.98	16.52	-1.04	17.50	15.87	32.49	8.49
	[2.35]	[2.93]	[-0.16]	[3.24]	[2.48]	[4.28]	[1.12]
umd	1.79	4.37	2.49	1.35	3.41	3.04	0.55
	[1.02]	[2.44]	[1.43]	[0.75]	[1.89]	[1.67]	[0.32]
# months	728	728	732	728	732	732	728
$\bar{R}^2$(%)	16	13	17	15	11	9	23

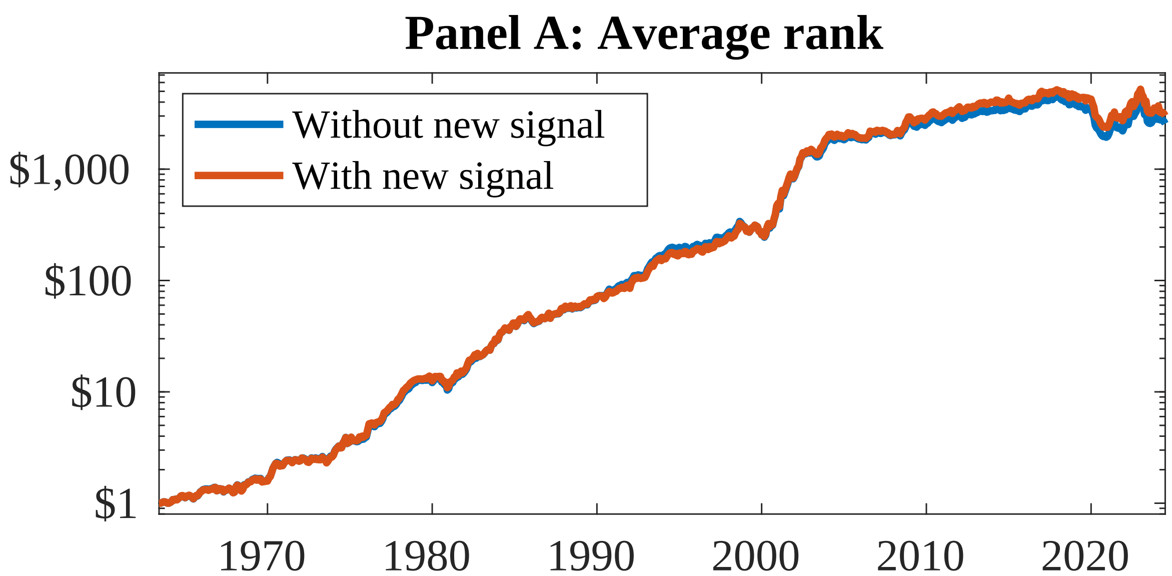


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ECD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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